



ELEC4150/8150 Electronic Design Proficiency

Project 2 – Signal Processing

INTRODUCTION

This project deals with audio signals, i.e. sound, which can be interfaced with a computer via its sound card, microphones, and speakers/headphones.

In this project, you are going to use software to design a series of signal processing functions for audio manipulation, transmission, and sound localisation effect. Your software is expected to import/record a song into your computer, analyse its spectrum/spectrogram, and separate the vocal and non-vocal components. Then, transmission will be simulated by compressing the resulted signal, followed by modulation, addition of noise, before being demodulated to recover the audio.

Your software is further expected to synthesize and reproduce sound localisation effect, which may be evaluated using a pair of headphones. The following website will give you a general sense about sound localisation:

<https://www.sony.com.au/electronics/360-reality-audio>

You can choose any programming language to complete this project, but will need to show all the code and explain how everything works.

PROJECT DESCRIPTION & SPECIFICATION

The project consists of the following parts:

1. To process audio signal, including design and implementation of a source separation algorithm to separate the vocal and the instrumental components.
2. To perform transmission of the audio signal at radio frequency over a noise channel, and to recover the audio at the receiver.
3. To simulate the signals received by multiple microphones with respect to the location of the sound source, and then to deduce the location of the source from the simulated signals.

Part 1 – Performing Audio Processing

- Import a piece of audio signal containing both vocal and instrumental components (e.g.

most songs), which can be either from a audio file or recorded using the microphone on your computer.

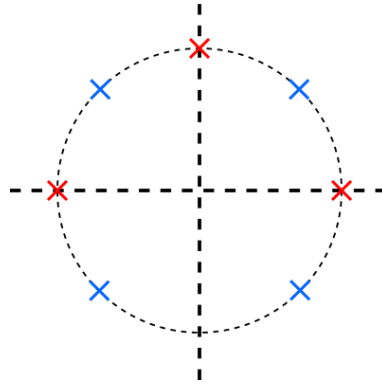
- Convert the sampling rate of the audio signal to 60 kHz, and display its spectrogram for 10 seconds in real time during playing.
- For the audio signal from the last step, design a method to extract the vocal component from the audio. Play the vocal component and illustrate how the spectrum has changed.
- Apply the method to extract the instrumental component and remove the vocal component as much as possible, producing the “karaoke” effect. While studio quality is not required, you should demonstrate that the vocal component have been attenuated by 40 dB or more.

Part 2 – Transmitting Signal over RF

- Compress the vocal signal obtained above in Part 1 to 64 kbps.
- Implement a modulation method of your choice (either Analogue or Digital) to modulate the compressed vocal signal onto a radio channel centred at 250 kHz, with maximum bandwidth of 4 kHz. Demonstrate how the spectrum of the vocal signal has been manipulated by your modulation method.
- Perform the corresponding demodulation on the modulated signal to obtain the transmitted vocal signal, and play it to validate successful transmission.
- The transmitted signal is assumed to be affected by an additive white Gaussian noise channel, resulting in signal-to-noise ratio of 10 dB. Include the noise in your simulation and demonstrate its influence on the transmitted and demodulated signals.
- Design and implement countermeasures that mitigate the interference from the additive white Gaussian noise as much as possible. Verify the performance of your countermeasures.

Part 3 – Simulating Sound Localisation

- In this part, you are expected to simulate the audio effect of sound localisation (that is, the sound comes from a certain location), based on the audio signals received by multiple microphones from the sound source. In particular, the microphones and the sound source are located in a horizontal plane (2D space). You can choose from two configurations for the microphones, shown in the diagram below, as either the red or the blue marks being placed on a circle of 1m radius. The sound source may be located randomly within the circle.



- Assuming the power of the audio signal reduces with the square of the distance from the source (i.e. following an inverse-square law), design a software system that takes the audio signal, the sampling rate, the fixed locations of multiple microphones, and the random location of the sound source as inputs, and outputs the signals received by each microphones. Demonstrate the output of your system using the vocal signal obtained in Part 1 as input.
- Because sampling of an audio signal contains non-negligible time delay from the sound source to the microphone, when multiple microphones are used, different time delays can be observed depending on each microphone's relative distance to the source. Assuming the speed of sound propagation is 343 m/s in dry air, modify your system to include relative time delays for each microphone corresponding to its distance to the sound source. Sub-sample delay resolution should be achieved as appropriate.
- Finally, assuming the relative time delay for each microphone can be measured perfectly, utilise the relative time delays in the simulated audio signals, together with the known geometric configuration of the microphones, to deduce the location of the sound source with respect to the microphones.

BONUS PART

The following functionality will gain bonus marks:

- For Part 1, create a note transcriber that maps the vocal component to music notes, including pitches and tempos.

THE DELIVERABLES

The total mark for the project consists of the following two deliverables:

1. Functioning simulation (which may contain multiple files) that meets the requirements of the project (25 marks). You will need to demonstrate the functions of your systems to the lab demonstrators and answer relevant questions during the practical class in Week 8.
2. A detailed project log/journal, containing the design process and your self-reflection on the experience undertaking the project (10 marks). The journal will require periodic inputs, and to be submitted via iLearn. It should include at least the following:
 - A journaled description of the progress of your project. This should include the rationale of your approach, resources that you have found (including relevant links), and successful as well as unsuccessful attempts on your design.
 - Any reflection of “lessons learned” for your unsuccessful attempts.
 - Things that you had to revise from your earlier knowledge in order to proceed.
 - References, including generative AI outputs and the prompts you used if applicable.
 - Schematics and drawings.

Assessment of the journal will be based on its content and also on your consistency in updating it periodically. You are required to bring your journal to the demonstration & defence in Week 8.

MARKING & ASSESSMENT

The following aspects will be marked:

- Your simulation, the design process undertaken, and how well you can demonstrate the effects of your systems.
- Your understanding and explanation of the operation of the systems from the signal processing perspective.
- To do this project properly you will have to do some mathematics, and you will be expected to explain your approach in your journal and during the oral defence.
- It is desirable to choose the signal processing techniques in such a way as to avoid unnecessary use of computational resources, while still meeting the specifications.
- How organised/neat your design is, and how easily you can troubleshoot it.

Details about the marking rubric will be provided in a separate document via iLearn.